Project: Worked examples
Title: Insulation thickness (ISO 12241)
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1. Scope and method

Calculates the insulation thickness of a pipe or flat surface that meets a heat-loss limit, a hot-surface limit or an anti-condensation requirement. For a chosen thickness it gives the heat flow, surface temperature and PASS or FAIL.

WHAT IT CALCULATES	RELATION
Pipe resistance per metre	$R = \ln(D_e/D_i)/(2\pi\cdot\lambda) + 1/(h_{se}\cdot\pi\cdot D_e)$, $D_e = D_i + 2t$; with the inner film and pipe wall $R_{oth} = 1/(h_i\cdot\pi\cdot D_p) + \ln(D_i/D_p)/(2\pi\cdot\lambda_w)$ added in series
Flat resistance per square metre	$R = t/\lambda + 1/h_{se}$ (+ $1/h_i + t_w/\lambda_w$)
Heat flow	$q = (\theta_i - \theta_a)/R$, positive from the process to the ambient
Surface temperature	$\theta_{se} = \theta_a + q/(h_{se}\cdot\pi\cdot D_e)$ (pipe), $\theta_a + q/h_{se}$ (flat)
Surface coefficient	h_{se} as given, or $h_{se} = h_c + h_r$ with $h_r = \varepsilon\cdot\sigma\cdot(T_s^2 + T_a^2)(T_s + T_a)$; h_c from free convection to air (laminar or turbulent, the larger), forced convection (cross-flow cylinder or flat plate) and their cube-law blend; solved with θ_{se}, which it depends on
Heat-loss limit	the thickness beyond which $\lvert q\rvert \le q_{max}$; surface-temperature limit (hot): $\theta_{se} \le \theta_{se,max}$; anti-condensation (cold): $\theta_{se} \ge \theta_{dew}$ + allowance, θ_{dew} entered (for example from the psychrometrics calculation) or from the Magnus relation at θ_a and RH
Thickness solution	each criterion's thickness by scanning t from 0 to 1 m for the last broken point and bisecting the bracket after it; governing t = max over the selected criteria
Critical diameter	$D_{crit} = 2\lambda/h_{se}$; below it, insulation raises the heat flow until D_e passes D_{crit}

Basis: Calculation rules for thermal insulation of equipment and installations (clause numbers to be checked)[1]; External forced, free and mixed convection correlations[2](§7.2, §7.4, §9.6, §9.9, 7th ed. (section numbers to be checked)); Magnus saturation vapour pressure (shared with the Glaser condensation calculation)[3](cl. 6 (clause number to be checked against the owner's copy)).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
geometry	Surface	pipe (cylinder)		assumption: A round pipe.	any
h_{mode}	Where the outer surface coefficient comes from	convection + radiation (correlation)		assumption: Surface coefficient computed from the correlation.	any
D_i	Pipe outer diameter, the inside of the insulation (pipe)	114	mm	drawing: M-PIP-021 rev B	10 … 1000 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
L_{surf}	Height or flow length of the flat surface (correlation, flat)	1	m	assumption: Not used: a pipe.	0.1 … 10 m
t_{chosen}	Chosen insulation thickness (optional; blank = report at the required thickness, with no checks; with it, nothing blocks)	50	mm	spec: Pipework schedule M-SCH-021: 50 mm sections	0 … 300 mm
λ	Thermal conductivity of the insulation	0.04	W/(m·K)	assumption: Mineral wool; take the maker's value at the mean temperature (W3 material)	0.015 … 0.2 W/(m·K)
T_{proc}	Process (fluid) temperature	120	°C	spec: SPEC-M-12 §3: heating flow temperature	-50 … 400 °C
T_{amb}	Ambient air temperature (also the radiation sink)	15	°C	spec: SPEC-M-12 §2: ambient design temperature	-20 … 50 °C
RH_a	Relative humidity of the ambient air (anti-condensation)	60	%	assumption: Not used: a hot pipe has no condensation criterion.	20 … 95 %
v_{wind}	Air speed over the surface (correlation; 0 = still air)	0	m/s	assumption: Still air, indoors.	0 … 20 m/s
h_{se}	Outer surface coefficient (as given; generic default of 10 W/(m²K), replace with the value for your surface; a high value is not conservative against condensation on a cold surface, use the correlation there)	10	W/(m²·K)	assumption: Not used: h_se is computed.	3 … 40 W/(m²·K)
eps	Emissivity of the outer surface (for the correlation; with h_{se} as given it only judges that value against condensation on a cold surface)	0.1		assumption: Bright aluminium cladding; replace with the cladding maker's emissivity.	0.05 … 1
$q_{max,m}$	Heat-loss limit per metre of pipe (optional; blank = not a criterion; the generic default of 25 W/m is to be replaced with your project limit)	60	W/m	spec: SPEC-M-12 §5: heat loss not to exceed 60 W/m	1 … 1000 W/m

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$\theta_{se,max}$	Highest surface temperature, for a hot surface (optional; blank = not a criterion)	60	°C	spec: SPEC-M-12 §6: personnel protection, surface not above 60 °C	30 ... 70 °C

3. Assumptions

A1. Heat flows in steady state, one-dimensionally, through a single homogeneous layer of constant conductivity. (Formula route of the standard.)
A2. The insulation has no thermal bridges at supports, no joints and no moisture. (Formula route of the standard.)
A3. The conductivity λ is the value entered, which should be the value at the mean insulation temperature on the sheet. (Stated.)
A4. With the correlation, the outer surface coefficient is $h_{se} = h_c + h_r$. (Incropera, §7.2, §7.4, §9.6 and §9.9 (section numbers to be checked against the owner's copy).)
A5. Radiation is $h_r = \varepsilon \cdot \sigma \cdot (T_s^2 + T_a^2)(T_s + T_a)$ to surroundings at the ambient air temperature. (Incropera, §1.2.)
A6. Free convection to air uses Holman's simplified relations at atmospheric pressure, the larger of the laminar and turbulent forms. (Holman's constants: horizontal cylinder 1.32(ΔT/D)^0.25 and 1.24ΔT^(1/3), vertical surface 1.42(ΔT/L)^0.25 and 1.31ΔT^(1/3).)
A7. Forced convection uses Churchill–Bernstein for a cylinder in cross-flow and the laminar or mixed flat-plate relations along a plate. (Incropera, §7.2 and §7.4.)
A8. Air properties are fits at the film temperature, with Pr = 0.71. (Stated.)
A9. Free and forced convection blend as $(h_{nat}^3 + h_{forced}^3)^{(1/3)}$. (Stated.)
A10. The correlation is indicative, not the standard's own. (The standard's own coefficients are not reproduced.)
A11. The dew point is entered, or is the Magnus relation at the ambient temperature and humidity. (Derived; the Glaser condensation calculation shares the function.)
A12. Below 0 °C the Magnus relation is over ice, so the humidity is then taken relative to ice. (Stated; enter the dew point for sub-zero air, because weather humidity is usually relative to water.)
A13. An allowance above the dew point is the user's choice. (Stated.)
A14. Pipes are treated as horizontal and flat surfaces as vertical for free convection. (Stated.)
A15. Air conductivity is a fit within about 3 % over 250 to 400 K. (Stated.)
A16. Air viscosity is a fit within about 4 % over 250 to 400 K. (Stated.)
A17. The required thickness is the least thickness beyond which the criterion holds for every thicker layer up to 1 m. (Stated; this matters when thin insulation on a small pipe raises the heat flow (critical diameter).)
A18. No thickness table of any standard, including BS 5422, is used or shipped. (No standard's table is reproduced.)
A19. The conductivity is the user's input, from a library material or the maker's data. (No standard's table is reproduced.)
A20. geometry (Surface) is taken as pipe (cylinder). (A round pipe.)
A21. h_{mode} (Where the outer surface coefficient comes from) is taken as convection + radiation (correlation). (Surface coefficient computed from the correlation.)
A22. L_{surf} (Height or flow length of the flat surface (correlation, flat)) is taken as 1 m. (Not used: a pipe.)
A23. λ (Thermal conductivity of the insulation) is taken as 0.04 W/(m·K). (Mineral wool; take the maker's value at the mean temperature (W3 material))
A24. RH_a (Relative humidity of the ambient air (anti-condensation)) is taken as 60 %. (Not used: a hot pipe has no condensation criterion.)
A25. v_{wind} (Air speed over the surface (correlation; 0 = still air)) is taken as 0 m/s. (Still air, indoors.)
A26. h_{se} (Outer surface coefficient (as given; generic default of 10 W/(m²K), replace with the value for your surface; a high value is not conservative against condensation on a cold surface, use the correlation

there)) is taken as 10 W/(m²·K). (Not used: h_{se} is computed.)
A27. eps (Emissivity of the outer surface (for the correlation; with h_{se} as given it only judges that value against condensation on a cold surface)) is taken as 0.1. (Bright aluminium cladding; replace with the cladding maker's emissivity.)

4. Working

[11] $t_{req,q}$ = t such that |q| ≤ 60 W/m for every thicker layer [1]
= scan 0 … 1000 mm, then bisection
= 23.374 mm
Required thickness, heat-loss limit

[12] $t_{req,se}$ = t such that θ_{se} ≤ 60 °C for every thicker layer [1]
= scan 0 … 1000 mm, then bisection
= 7.8423 mm
Required thickness, surface-temperature limit

[13] thickness = max over the selected criteria [1]
= $t_{req,q}$ = 23.374 mm, $t_{req,se}$ = 7.8423 mm
= 23.374 mm
Governing required thickness (heat-loss limit)

[15] $D_e = D_i + 2t$
= 114 mm + 2 × 50 mm
= 214 mm
Outer diameter of the insulation, chosen thickness

[16] $R_{ins} = \ln(D_e/D_i)/(2\pi\cdot\lambda)$ [1]
= ln(214/114)/(2π × 0.04)
= 2.5058 m·K/W
Resistance of the insulation

[17] h_c = free convection to air, blended with forced convection [2](7th ed. §7, §9.6, §9.9)
= horizontal cylinder, Holman; Churchill–Bernstein if the wind blows
= 3.6748 W/(m²·K)
Convective surface coefficient at the surface temperature

[18] $h_r = \varepsilon\cdot\sigma\cdot(T_s^2 + T_a^2)(T_s + T_a)$ [2](7th ed. §1.2)
= ε = 0.1, θ_{se} = 27.855 °C
= 0.58007 W/(m²·K)
Radiative surface coefficient

[19] $h_{se} = h_c + h_r$ [1]
= 3.6748 + 0.58007
= 4.2549 W/(m²·K)
Outer surface coefficient used

[20] $R_{se} = 1/(h_{se}\cdot\pi\cdot D_e)$ [1]
= 1/(4.2549 × π × 0.214 m)
= 0.34958 m·K/W
Resistance of the outer surface

[21] $R_{tot} = R_{oth} + R_{ins} + R_{se}$ [1]
= 0 + 2.5058 + 0.34958
= 2.8554 m·K/W
Total resistance per metre

[22] $q_{per,m} = |\theta_i - \theta_a| / R$ [1]
= 105 K / 2.8554
= 36.773 W/m
Heat flow per metre of pipe (heat loss, magnitude)

[23] $T_{se} = \theta_a + q/(h_{se}\cdot\pi\cdot D_e)$ [1]
= 15 °C + 36.773 / 2.8606
= 27.855 degC
Outer surface temperature

[24] $T_m = (\theta_i + \theta_{se})/2$
$= (120 + 27.855)/2$
$= 73.927$ degC
Mean temperature of the insulation: take λ at this temperature
[25] $D_{crit} = 2\lambda/h_{se}$ at the bare surface [1]
$= 2 \times 0.04 / 8.1897$
$= 9.7683$ mm
Critical diameter at the bare-surface h_{se}: below it, insulation raises the heat flow
[26] $u_q = |q| / q_{max}$
$= |q| \le 60$ W/m
$= 0.61288$
Utilisation, heat-loss limit
[27] $u_{se} = (\theta_{se} - \theta_a)/(\theta_{se,max} - \theta_a)$
$= \theta_{se} \le 60$ °C
$= 0.28567$
Utilisation, surface-temperature limit
[28] $dT_{se} = \theta_{se} - \theta_a$
$= 27.855$ °C − 15 °C
$= 12.855$ K
Rise of the surface above the ambient air
[29] $dT_{se,max} = \theta_{se,max} - \theta_a$
$= 60$ °C − 15 °C
$= 45$ K
Largest rise of the surface above the ambient air

5. Results

SYMBOL	VALUE	DESCRIPTION
thickness	23.37 mm	Governing required thickness
$q_{per,m}$	36.77 W/m	Heat flow per metre (heat loss)
T_{se}	27.85 °C	Outer surface temperature
$t_{req,q}$	23.37 mm	Required thickness, heat-loss limit
$t_{req,se}$	7.842 mm	Required thickness, surface-temperature limit
T_m	73.93 °C	Mean insulation temperature
R_{tot}	2.855 m·K/W	Total resistance
D_{crit}	9.768 mm	Critical diameter (at the bare-surface h_{se})
h_c	3.675 W/(m²·K)	Convective surface coefficient
h_r	0.5801 W/(m²·K)	Radiative surface coefficient
h_{se}	4.255 W/(m²·K)	Outer surface coefficient
governing	"heat-loss limit"	Governing criterion
u_q	0.61	Utilisation, heat-loss limit (governing)
u_{se}	0.29	Utilisation, surface-temperature limit

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Chosen thickness against the heat-loss limit	36.77 W/m	60 W/m	0.61	[1]
PASS	Chosen thickness against the surface-temperature limit	12.85 K	45 K	0.29	[1]

7. Applicability

WARNING records: tabulated values not taken from a reference record are λ (Material properties); h_{se} (Outer surface coefficient); eps (Surface emissivity); $q_{max,m}$, $\theta_{se,max}$ (Insulation criteria). Enter each as one so its source, edition and licence travel with it. [record]

8. Verification

```
Template bs.insulation@1.2.0 Content hash b8b77970e70183e1 Engine calc-engine/1.0.0
Known-answer cases passed 14/14 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. ISO 12241:2022.
2. Bergman TL, Lavine AS, Incropera FP, DeWitt DP. Fundamentals of Heat and Mass Transfer. 7th ed. Wiley; 2011.
3. BS EN ISO 13788:2012.